Software Requirements Specification (SRS)

Traffic Management System (Simulation)

Field	Value
Project	Traffic Management System (Simulation) — a microscopic road-traffic simulator for designing and evaluating traffic-signal control strategies
Course	Software Engineering UE24CS341A
Institution	PESU
Version	0.2 (Simplified draft)
Date	2026-09-07
Status	Final submission
Team	4 members (see Authors below)

1. Introduction

1.1 Purpose

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Revision history

Version	Date	Author	Change summary	Approval
0.2	2026-09-07	Venkata Srivaibhav Nukaraju V Vyas Vikas N Naik Vikhyath Bharadwaj K S	Finalized simplified SRS for implementation and review.	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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TMS-Sim is a software-only desktop application. It lets a traffic analyst build a road network, configure fixed-time traffic signals, describe traffic demand, and run a time-stepped microscopic simulation of individual vehicles moving through that network. The tool reports measures such as delay, queue length, throughput, speed and travel time so that signal plans can be compared on the same scenario.

In scope:

- Interactive modelling of intersections, road segments, lanes and turning movements.
- Fixed-time traffic-signal control at signalized intersections.
- Seedable, reproducible vehicle generation from entry-point arrival rates and turning ratios.
- Time-stepped simulation with a documented car-following and gap-acceptance model.
- 2D visualization, run metrics, comparison of completed runs and CSV export.
- A local scenario library for saving and loading scenarios and completed runs.

Out of scope:

- Controlling or connecting to real-world traffic signals, sensors or field hardware.
- 3D visualization, distributed/multi-machine simulation, and advanced lane-changing behaviour.
- Calibration against real field-measured traffic data; correctness is defined against the documented model equations.
- Public transport scheduling, pedestrian and cyclist micro-simulation.

1.3 Audience

Development team (4 members), Project Guide, Course Coordinator and Evaluators, and any future student maintainers. Secondary readers: QA/testers and users acting as traffic analysts during acceptance testing.

1.4 Definitions and acronyms

Term	Meaning
TMS-Sim	Traffic Management System (Simulation) — the product specified here
SRS	Software Requirements Specification
FR / NFR	Functional Requirement / Non-Functional Requirement
RTM	Requirements Traceability Matrix
Car-following model	Rule set (e.g. Intelligent Driver Model) giving a vehicle's acceleration from the gap and speed of the vehicle ahead
Gap acceptance	Rule deciding whether a vehicle enters a conflicting movement given the available time gap
Phase / Cycle	A phase is an interval during which a set of non-conflicting movements gets green; the cycle is one full run through all phases
MOE	Measure of Effectiveness: delay, queue length, throughput, stops, travel time, average speed
Delay	Extra travel time a vehicle experiences versus free-flow travel over the same path

Throughput	Number of vehicles that complete their assigned route within the simulated period
PRNG / Seed	Pseudo-Random Number Generator / the integer that makes a random stream reproducible
FPS	Frames Per Second (visualization refresh rate)
Real-time factor	Simulated seconds advanced per wall-clock second (10× = ten simulated seconds per real second)
CSV / JSON	Comma-Separated Values / JavaScript Object Notation file formats
Reference hardware	The agreed baseline machine for performance figures: 4-core Intel i5-11400H @ 2.70 GHz, 16 GB RAM

1.5 References

- IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications.
- Course material: “Requirements Engineering: Software Requirement Specification, RTM”, Unit 1, Lecture 9–10 (UE24CS341A).
- Course SRS template: “SRS_Template for SE Reference.docx”.
- Treiber, Hennecke, Helbing (2000), “Congested traffic states in empirical observations and microscopic simulations” — Intelligent Driver Model (reference model).

2. Overall description

2.1 Product perspective

TMS-Sim is a self-contained desktop application with a Simulation Engine, Network/Signal Editor, Demand Generator, 2D Renderer, Metrics service and Scenario Store. The system runs locally on a single computer.

2.2 Major product functions

- Provide a desktop workbench for the traffic analyst.
- Build and validate a road network of intersections, segments, lanes and turning movements.
- Configure fixed-time signal plans and validate them for safety and timing.
- Define traffic demand using entry-point arrival rates and turning ratios for passenger cars.

Role	Characteristics & needs
Analyst (primary user)	Primary user. Builds networks, configures signals and demand, runs simulations, and interprets results.
Simulation Engine (system actor)	Internal component that executes the model and produces simulation state and metrics.

2.4 Operating environment

- Desktop build: Windows 10/11; keyboard, mouse/trackpad, display ≥ 1366×768.
- Implementation stack: Python 3.11 + PySide6
- Local storage: SQLite database or a scenarios folder on disk.
- No internet connection is required to model or run simulations.

2.5 Constraints

- Academic project: one semester, team of 4, developed and demonstrated on student laptops.
- The microscopic model is limited to one documented car-following model plus a simple gap-acceptance rule; no cooperative lane-changing in v1.
- 2D visualization only; single-machine execution (no distributed simulation).
- Signal plans must never allow conflicting movements to be green together (safety constraint enforced by validation).
- Random behaviour must be driven by a single seedable PRNG so results are reproducible.
- Third-party libraries must be available under permissive open-source licenses.

2.6 Assumptions and dependencies

- Users supply their own network data or start from the bundled sample scenarios.
- Performance figures are quoted against the agreed reference hardware defined in §1.4.
- Required Python libraries are available under permissive open-source licenses.
- Correctness is judged against the documented model equations, not against real-world field measurements.

3. External interface requirements

3.1 User interfaces

- A single workbench window with: a network canvas (centre), a properties/inspector panel (right), a palette/toolbox (left), a simulation control bar (start/pause/step/reset, speed multiplier, clock) and a results dashboard (dockable).
- Network editing by direct manipulation: click to place intersections, drag to draw segments, select to edit properties.
- Validation errors shown in a list; clicking an error highlights the offending element on the canvas.
- Results dashboard shows live time-series charts during a run and a summary table plus charts after it.
- Usability — the main workflow is understandable and validation messages are clear.

3.2 Hardware interfaces

None beyond a standard personal computer (CPU, RAM, display, keyboard, pointing device). TMS-Sim does not interface with sensors, controllers or any traffic field hardware. A discrete GPU is optional and only improves rendering of large networks.

3.3 Software interfaces

Interface	Description
Operating system	Windows desktop environment providing file dialogs and windowing.
Scenario store	SQLite database or JSON files on the local filesystem for scenarios and completed runs.
Scenario file format	Import/export of a full scenario as one JSON file conforming to a published schema (version-tagged).
CSV export	Summary metrics from completed runs can be exported as CSV files.

3.4 Communications

- No network communication is required.

The remainder of this SRS defines the functional and non-functional requirements, each with acceptance criteria and a traceable test-case reference.

4. System features (detailed functional requirements)

4.1 Road network modelling

Let the Analyst build and validate the road network the simulation runs on.

Req ID	Requirement (the system shall…)	Priority	Acceptance criteria / Test case ref	Source & dependencies
TMS-F-010	shall let an Analyst create and edit a road network as a directed graph of intersections and road segments, each segment having length, lane count, direction and speed limit.	High	AC-010: a saved network reloads with every element and attribute intact. Test: TC-NET-01	Analyst
TMS-F-011	shall let the Analyst mark each intersection as signalized, stop-controlled, yield-controlled or uncontrolled, and define the permitted turning movements for every approach.	High	AC-011: only defined turning movements are available to vehicles during a run. Test: TC-NET-02	Analyst
TMS-F-012	shall validate the network before a run and block the run if the graph is disconnected, a segment has zero lanes or zero length, or a turning movement has no receiving lane — listing every error.	High	AC-012: each seeded defect produces a specific, located error and the Run action stays disabled. Test: TC-NET-03	Quality; safety
TMS-F-013	shall import and export a complete network (and its signal plans and demand) as a single version-tagged JSON scenario file conforming to the published schema.	Medium	AC-013: export then import on a clean install reproduces an identical scenario. Test: TC-NET-04	Interoperability

4.2 Traffic-signal configuration

Let the Analyst define and validate a fixed-time signal-control plan.

Req ID	Requirement (the system shall…)	Priority	Acceptance criteria / Test case ref	Source & dependencies

TMS-F-020	shall let the Analyst define, per signalized intersection, an ordered set of phases with green, yellow and all-red durations.	High	AC-020: the editor shows each phase and its timing values. Test: TC-SIG-01	Analyst
TMS-F-022	shall validate a signal plan and reject it if phase timings are invalid or conflicting movements are green at the same time.	High	AC-022: each seeded timing/conflict error is reported and the plan cannot be used. Test: TC-SIG-02	Safety constraint

4.3 Traffic demand

Describe how much traffic enters the network.

Req ID	Requirement (the system shall…)	Priority	Acceptance criteria / Test case ref	Source & dependencies
TMS-F-030	shall let the Analyst specify entry-point arrival rates (vehicles/hour) and turning ratios.	High	AC-030: generated entry counts are reasonably consistent with the configured rates. Test: TC-DEM-01	Analyst
TMS-F-032	shall use a single seedable pseudo-random stream so that two runs with the same scenario and seed produce the same vehicle arrivals and metrics.	High	AC-032: same scenario + seed produces identical results. Test: TC-DEM-02	Reproducibility (quality attribute)

4.4 Simulation execution & control

Run the microscopic simulation and let the Analyst control it.

Req ID	Requirement (the system shall…)	Priority	Acceptance criteria / Test case ref	Source & dependencies
TMS-F-040	shall advance the simulation in fixed time steps and update each vehicle using the documented car-following model and a simple gap-acceptance rule.	High	AC-040: vehicles follow the documented rules and do not cross a red signal. Test: TC-SIM-01	Core objective
TMS-F-041	shall provide start, pause, single-step, reset and speed controls with 0.5×, 1×, 2× and 4× options.	High	AC-041: pause freezes the simulation, single-step advances one step, and reset returns to t = 0. Test: TC-SIM-02	Analyst
TMS-F-042	shall let the Analyst set a total simulated duration and stop the run automatically when it is reached, finalising all metrics.	High	AC-042: the run ends at the configured simulated time ± one time step and metrics are written once. Test: TC-SIM-03	Analyst
TMS-F-043	shall render a 2D top-down view of the network showing vehicle positions, signal indications and queues, with pan and zoom.	High	AC-043: displayed vehicle positions and signals match engine state. Test: TC-SIM-04	Analyst

4.5 Metrics, results & reporting

Turn a run into measures the Analyst can compare and export.

Req ID	Requirement (the system shall…)	Priority	Acceptance criteria / Test case ref	Source & dependencies

TMS-F-050	shall compute average and maximum vehicle delay, average queue length per approach, throughput, average speed and average travel time for each completed run.	High	AC-050: computed values match an independent recomputation from the event log within rounding. Test: TC-MET-01	Core objective
TMS-F-051	shall display key metrics during a run and a summary of results after the run.	Medium	AC-051: displayed values update during the run and match final metrics. Test: TC-MET-02	Analyst
TMS-F-052	shall persist each completed run with its scenario, random seed, configuration and computed metrics.	High	AC-052: a saved run can be reloaded and displayed. Test: TC-MET-03	Analyst
TMS-F-053	shall let the Analyst select two completed runs and compare their metrics in a table.	Medium	AC-053: the table shows the metric values and differences for both runs. Test: TC-MET-04	Core objective
TMS-F-054	shall export a run's summary metrics as a CSV file.	Medium	AC-054: exported CSV contains headers and the displayed metrics. Test: TC-MET-05	Analyst

4.6 Scenario library

Store scenarios and completed runs.

Req ID	Requirement (the system shall…)	Priority	Acceptance criteria / Test case ref	Source & dependencies
TMS-F-060	shall let the Analyst save, list, load,	High	AC-060: each operation is reflected in the	Analyst

	rename and delete named scenarios.		scenario list and survives a restart. Test: TC-LIB-01	

5. Non-functional requirements

NFRs are measurable and tied to a test in the acceptance suite. IDs follow TMS-NF-###. Performance figures are quoted against the reference hardware in §1.4.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
TMS-NF-001	On the reference hardware, the system shall run a scenario with up to 10 intersections and 500 vehicles without crashing and shall maintain at least 15 FPS during normal visualization.	Performance	High	A 1-hour test run completes successfully and the average visualization rate is at least 15 FPS. Test: TC-PERF-01
TMS-NF-002	Reliability	5 / DS-QA-02	High	TC-REL-01
TMS-NF-003	Usability	5 / DS-QA-03	Medium	TC-UX-01
TMS-NF-004	Portability	5 / DS-QA-04	Medium	TC-PORT-01

Reproducibility is treated as a quality attribute and is verified through TMS-F-032 (same scenario + seed → identical metrics).

6. Quality attributes & Acceptance tests

6.1 Key quality attributes

- Correctness & reproducibility — model behaviour follows the documented equations; identical scenario and seed produce identical results.
- Performance — the application runs the reference scenario without crashing and maintains acceptable visualization performance.
- Usability — the main workflow is understandable and validation messages are clear.
- Portability — the desktop application runs on Windows 10/11.
- Maintainability — simulation, UI, storage and metrics are separated into modules.
- Testability — the simulation engine can be tested independently of the visualization.

6.2 Exit criteria for acceptance

- Every High-priority functional requirement is implemented and its acceptance test passes.

- All 4 non-functional requirements are checked and their acceptance criteria are met.
- The reproducibility check (TMS-F-032) passes.
- The RTM in §8 shows every retained requirement linked to at least one test case.
- No open defect of severity Critical or High.

6.3 Acceptance test suites

Suite	Covers
Network Modelling	TMS-F-010 … TMS-F-013
Signal Configuration	TMS-F-020 … TMS-F-022
Demand Generation	TMS-F-030 … TMS-F-032
Simulation Engine & Controls	TMS-F-040 … TMS-F-043
Visualization	TMS-F-043
Metrics & Reporting	TMS-F-050 … TMS-F-054
Scenario Library	TMS-F-060
Performance / Reliability	TMS-NF-001 … TMS-NF-002
Usability / Portability	TMS-NF-003 … TMS-NF-004

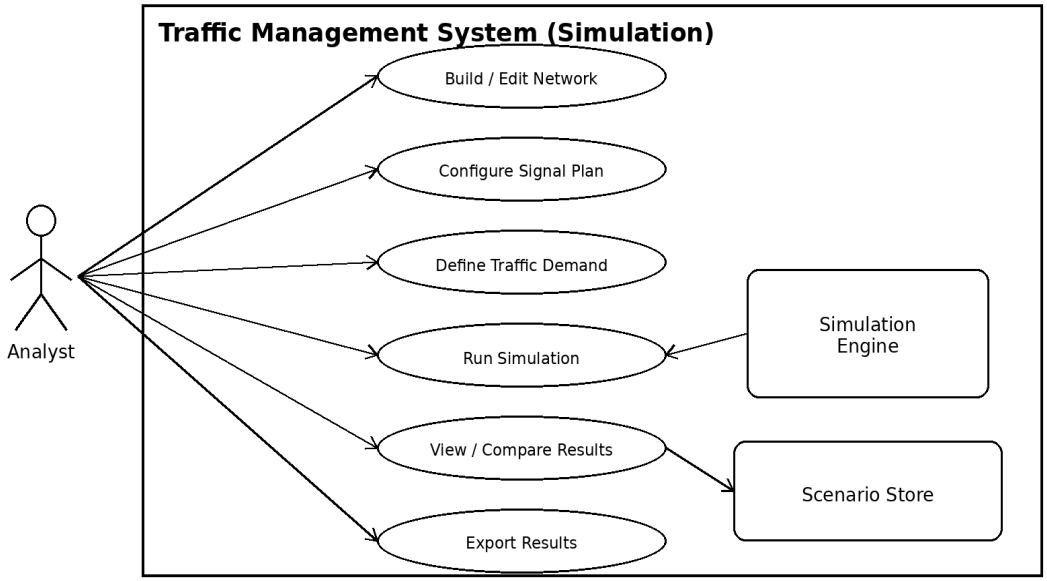
7. System models and diagrams

7.1 Use-case diagram — System context

Actor: Analyst (primary user). The Simulation Engine is an internal component.

Use cases: Build / Edit Network, Configure Signal Plan, Define Traffic Demand, Run Simulation, View Results, Compare Runs, Export Results, Manage Scenario Library.

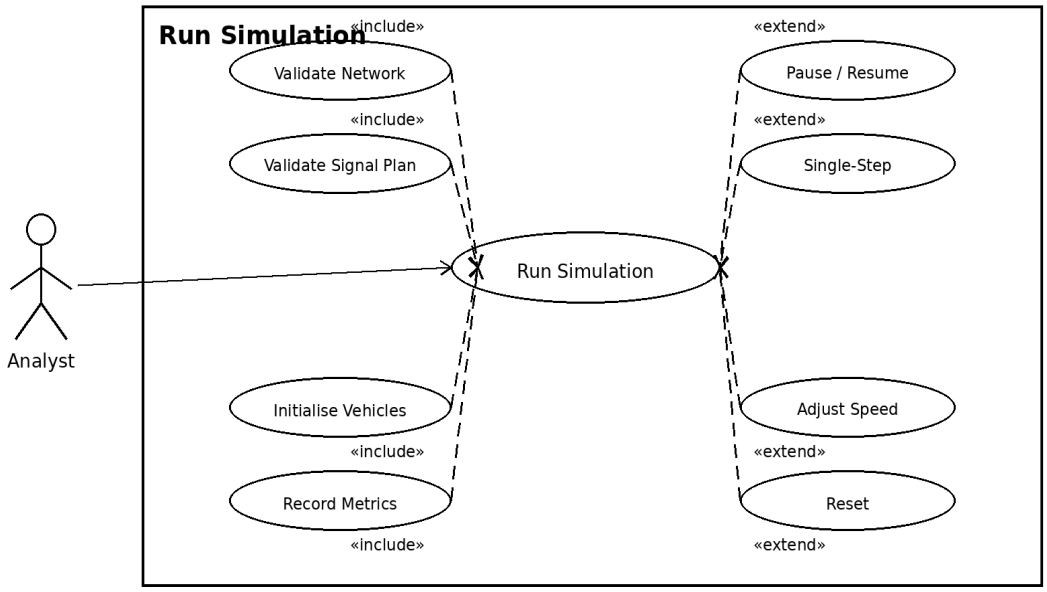
Use-Case Diagram — System Context



7.2 Use-case diagram — “Run Simulation” decomposition

Actor: Analyst (primary user). Included activities: Validate Network, Validate Signal Plan, Initialise Vehicles, Run Simulation, Record Metrics. Optional controls: Pause, Single-Step, Adjust Speed, Reset.

Use-Case Diagram — “Run Simulation” Decomposition



8. Requirements Traceability Matrix (RTM)

Status legend: N = Not started, P = Partially verified, A = Accepted (test passed). Design-spec and test-case IDs will be completed during implementation and testing.

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status	Comments
TMS-F-010	let an Analyst create and edit a road network as a directed …	4.1 / DS-NET-01	NetworkEditor	TC-NET-01	N	
TMS-F-011	let the Analyst mark each intersection as signalized	4.1 / DS-NET-02	NetworkEditor	TC-NET-02	N	
TMS-F-012	validate the network before a run and block the run if the g…	4.1 / DS-NET-03	NetworkEditor	TC-NET-03	N	
TMS-F-013	import and export a complete network (and its signal plans a…	4.1 / DS-NET-04	NetworkEditor	TC-NET-04	N	
TMS-F-020	let the Analyst define fixed-time signal phases	4.2 / DS-SIG-01	SignalConfig	TC-SIG-01	N	
TMS-F-022	validate a signal plan	4.2 / DS-SIG-03	SignalConfig	TC-SIG-02	N	
TMS-F-030	specify entry-point arrival rates and turning ratios	4.3 / DS-DEM-01	DemandGenerator	TC-DEM-01	N	
TMS-F-032	generate vehicles from a seedable pseudo-random stream	4.3 / DS-DEM-03	DemandGenerator	TC-DEM-02	N	

TMS-F-040	advance the simulation in fixed time steps	4.4 / DS-SIM-01	SimulationEngine	TC-SIM-01	N	
TMS-F-041	provide simulation controls	4.4 / DS-SIM-02	SimulationEngine	TC-SIM-02	N	
TMS-F-042	let the Analyst set a total simulated duration and stop the …	4.4 / DS-SIM-03	SimulationEngine	TC-SIM-03	N	
TMS-F-043	render a real-time 2D view	4.4 / DS-REN-01	Renderer	TC-SIM-04	N	
TMS-F-050	compute run metrics	4.5 / DS-MET-01	MetricsService	TC-MET-01	N	
TMS-F-051	display key metrics during and after a run	4.5 / DS-MET-02	MetricsService	TC-MET-02	N	
TMS-F-052	persist completed runs	4.5 / DS-LIB-02	ScenarioStore	TC-MET-03	N	
TMS-F-053	compare two completed runs	4.5 / DS-MET-03	MetricsService	TC-MET-04	N	
TMS-F-054	export summary metrics as CSV	4.5 / DS-MET-04	ReportExporter	TC-MET-05	N	
TMS-F-060	save, list, load, rename and delete scenarios	4.6 / DS-LIB-01	ScenarioStore	TC-LIB-01	N	
TMS-NF-001	Performance	5 / DS-QA-01	Cross-cutting	TC-PERF-01	N	
TMS-NF-002	Reliability	5 / DS-QA-02	Cross-cutting	TC-REL-01	N	

TMS-NF-003	Usability	5 / DS-QA-03	Cross-cutting	TC-UX-01	N	
TMS-NF-004	Portability	5 / DS-QA-04	Cross-cutting	TC-PORT-01	N	